Cherry Diseases

Identification, causes, favourable conditions, management and verified remedies

Six major diseases of cherry (Prunus avium), temperate-India focus (Jammu & Kashmir, Himachal Pradesh, Uttarakhand)

Primary sources: Government of India NIPHM-DPPQS AESA IPM package for cherry; CIB&RC Major Uses of Pesticides - Fungicides (up to 30/09/2025); UC IPM and other university extension material.

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About this document

Scope and method

Each disease chapter follows the same six headings: identification; causes and progression; favourable conditions; prevention and management; remedies and treatment; and sources. The content was reproduced from the supplied cherry reference and re-typeset in the common layout used for the other crops. Chemical recommendations are included only where they could be matched to the CIB&RC compilation of registered uses on cherry (V1) or to the Government of India IPM package (V2).

Verification tags used in the chemical tables

V1 - Verified against CIB&RC. The product is listed for CHERRY and this disease in the CIB&RC 'Major Uses of Pesticides - Fungicides' compilation (up to 30/09/2025). The dose is reproduced as printed there.

V2 - Government / university recommendation only. Published in the Government of India NIPHM-DPPQS AESA IPM package for cherry (N1). The cherry label claim for this disease could NOT be matched in the CIB&RC compilation, so confirm the current CIB&RC registration and product label before use.

-- - Not recommended / no chemical option verified. The sources reviewed do not support a chemical recommendation for cherry in India, so none is given.

Important cautions

- The CIB&RC compilation used for V1 is dated up to 30/09/2025 and was read from a copy hosted on a university website, not from the live CIB&RC register. Registrations, label claims and permitted uses change. Always check the current label and the live register (cibrc.gov.in) and follow the pre-harvest interval (PHI) printed on the label.
- CIB&RC states that its compilation is for guidance and not for legal purposes. Where the entry for fruit trees gives a concentration and a volume per tree, that is how it is reproduced here.
- Doses are 'per hectare (ha)' where taken from CIB&RC and 'per acre (ac)' where taken from the Indian IPM package (N1). 1 ha = 2.47 ac. Water volumes depend on tree size, stage and sprayer type.
- Rotate fungicide groups across sprays to slow resistance. Use PPE, follow label safety instructions and keep products away from water bodies.
- Most of the disease descriptions come from the Government of India IPM package for cherry (N1) and from international university extension pages, because little Indian cherry-specific disease material was retrievable. Where an international source is used, it is tagged and, for biology and cultural practice only, not as an Indian spray recommendation.
- No SKUAST-K, YSPUHF Nauni, ICAR-CITH, KVK, TNAU, MPKV Rahuri or PDKV Akola spray schedule for cherry diseases was retrieved during this review, so none is included.

Summary table

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Brown Rot	Monilinia fructicola and Monilinia laxa (Fungal (mummy- and canker-borne; airborne spores))	Sanitation (mummies, cankers, fallen fruit); avoid fruit wounds; resistant varieties where available	V1 options (captan 50% WP, 75% WP)
Bacterial Canker (Gummosis)	Pseudomonas syringae (pv. syringae) (Bacterial (plant-surface-, rain- and wound-borne))	Healthy stock; keep water off canopy; avoid late nitrogen; correct pruning time; avoid winter injury	None verified (tag --)
Cherry Leaf Spot (Shot Hole)	Blumeriella jaapii in the international literature; Cercospora circumcissa / C. rubrotincta in the Indian IPM package (Fungal (leaf-debris- and soil-borne))	Destroy fallen leaves; balanced N and K; good air movement; avoid overhead wetting	V1 options (lime sulphur, zineb); captan V2
Powdery Mildew	Podosphaera clandestina (Fungal (obligate parasite; airborne conidia))	Spacing and pruning for air flow; remove infected shoots and suckers; early scouting	None verified (tag --)
Crown Gall	Agrobacterium tumefaciens (also named Rhizobium radiobacter) (Bacterial (soil- and nursery-stock-borne; wound entry))	Pathogen-free nursery stock; avoid wounding; avoid infested soil; destroy galled trees	None verified (tag --)
Collar Rot / Root and Crown Rot (Phytophthora)	Phytophthora spp. (Oomycete (fungus-like), soil- and water-borne)	Well-drained site; berms; no sprinkler wetting of crown; short frequent irrigations	None verified (tag --)

Quick separation of look-alikes: blossoms and twigs that wither tan-brown with grey spore masses and later brown, mummified fruit point to brown rot; amber gum on cankers, black-killed shoots and dead buds point to bacterial canker; spots with reddish-brown margins that drop out as shot-holes, with early leaf fall, point to leaf spot; white powdery growth under the leaf and on shoot tips points to powdery mildew; soft spongy galls at the crown or roots point to crown gall; and sudden wilting after the first warm spring weather with dark bark at the crown in wet soil points to Phytophthora collar rot.

1. Brown Rot

Cherry | Monilinia fructicola and Monilinia laxa

Crop	Cherry (Prunus avium)
Disease	Brown Rot
Pathogen	Monilinia fructicola and Monilinia laxa
Type	Fungal (mummy- and canker-borne; airborne spores)

At a glance. Blossom, twig and fruit disease of stone fruits. The fruit-rot phase does the most damage: ripening fruit can rot fast in warm, wet, humid weather. Sanitation, blossom protection and avoiding fruit wounds matter most.

1. Identification

Symptoms

- Crop / stage: blossoms, spurs, twigs and ripening fruit of sweet cherry.
- Blossoms: infected flowers turn brown and wither, and either stay fixed to the twig as a gummy mass or drop like unpollinated flowers (MB).
- Twigs and leaves: if blighted flowers stay on, the fungus can grow into the twig below and cause cankers with gumming at the edge of healthy tissue; leaves on infected shoots turn brown and wither but stay attached (MB).
- Fruit: soft brown spots that expand quickly and become covered with powdery tan spore masses; the rot spreads between fruits in contact, and infected fruit often stays on the tree as mummies (MB).
- Rot can develop on the tree or after harvest in storage; mature fruit are the most susceptible (UCH).

Distinguishing features

- Blossom blight can be confused with blossom blast caused by Pseudomonas syringae. Shoots killed by brown rot turn mostly brown to tan, whereas shoots killed by bacterial blast generally turn black (UCH, MB).
- Brown rot fruit carries powdery grey to light-brown spores; Alternaria rot carries dark green to brown spore masses; Rhizopus rot is softer (MB). The Indian IPM package lists Rhizopus 'whisker rot' as a separate disease (N1).
- European brown rot (M. laxa) is described as far more aggressive on blossoms and spurs than American brown rot (M. fructicola) (NEB).

2. Causes and progression

- Pathogens: Monilinia fructicola and Monilinia laxa (UCD). The Indian IPM package groups 'leaf spot / shot hole / brown rot' in one entry without a separate Monilinia description (N1), so the pathogen description here rests on university sources.
- Survival: mainly in fruit mummies, peduncles and cankers (NEB). On peach, mummies that fall to the ground form apothecia in spring that release ascospores at the same time as bloom, and these are the primary inoculum in many orchards (UCB).
- Spread: spores are airborne or rain-splashed (UCR).
- Progression: blossom infection leads to blossom blight and twig cankers; later, spores landing on injured or moist fruit cause brown or tan circular spots that enlarge (UCH). Fruit can be infected through the cuticle, but more easily through natural openings and wounds; rain-cracked fruit are especially prone, and the amount of inoculum drives how severe infection becomes (NEB).

3. Favourable conditions

- Warm, wet or humid weather (MPR). Little or no blossom blight develops if blossoms do not become wet during bloom (UCH).
- Wounding, fruit cracking and moisture increase the likelihood of infection (UCD). Fruit become more susceptible as they mature (NEB).

- Heavy carry-over inoculum from mummies and cankers (NEB). Under conducive conditions the whole fruit may rot within about 48 hours (MPR).
- Sweet cherries are generally more susceptible than tart cherries (NEB).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Sanitation: remove and destroy mummies, cankered twigs and fallen fruit. The Indian IPM package advises collecting and destroying fallen fruits (N1); sanitation is described as essential for brown rot control (CT).
- Manage other diseases and pests that injure fruit, because wounds raise the chance of brown rot infection (UCH).
- Choose resistant varieties where available (UCH). No brown-rot-resistant cherry cultivar for Indian conditions was found in the sources reviewed; the IPM package refers growers to the nearest KVK / SAU / ICAR institute for resistant or tolerant varieties (N1).
- After harvest, prompt cooling and refrigerated storage reduce postharvest losses (UCD).

Regional note (temperate India)

No state-specific brown rot recommendation (SKUAST-K, YSPUHF Nauni, ICAR-CITH) was found in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Captan 50% WP	Cherry - brown rot	2.5 kg formulation/ha (1250 g a.i./ha)	750-1000 L/ha	V1	CIB&RC; waiting period not stated in the entry. Cross-check (N1): 1 kg per acre in 300-400 L water.
Captan 75% WP	Cherry - brown rot	0.12% a.i. in the spray solution (1667 g formulation as printed)	15-20 L per tree	V1	CIB&RC; waiting period 'NA'. Fruit-tree entry: values are spray-solution concentration and volume per tree.

V1 - Verified against CIB&RC. The product is listed for CHERRY and this disease in the CIB&RC 'Major Uses of Pesticides - Fungicides' compilation (up to 30/09/2025). The dose is reproduced as printed there.

- Do not mix captan with lime sulphur: alkaline materials such as lime sulphur reduce captan's fungicidal activity, and captan should not be combined with, or applied just before or after, oil sprays (EPA label statement, US).
- Leaf injury from captan has been reported on some US sweet cherry cultivars when applied between petal fall and harvest (CT). Behaviour on cherry cultivars grown in India is not documented in the sources reviewed.
- Peach guidance links bloom-time protection to less ripe-fruit rot at harvest (UCB). The CIB&RC entries do not specify a timing, so follow the product label.
- No other fungicide could be matched to cherry brown rot in the CIB&RC compilation. Products used on cherry abroad (for example sterol-inhibitor, strobilurin or dicarboximide groups) are not reproduced because no cherry registration for them was found.

6. Sources and citations

[N1] AESA based IPM package for Cherry. Directorate of Plant Protection, Quarantine & Storage (DPPQS) and National Institute of Plant Health Management (NIPHM), Ministry of Agriculture & Farmers Welfare, Government of India. https://niphm.gov.in/IPMPackages/Cherry.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS - Major Uses of Pesticides: Fungicides (registered under the Insecticides Act, 1968), up to 30/09/2025. Copy reviewed: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[UCD] UC Davis Fruit & Nut Research and Information Center - Cherry diseases and disorders. https://fruitsandnuts.ucdavis.edu/node/5006

[UCH] UC IPM - Brown rot blossom and twig blight (home and landscape; stone fruit). https://ipm.ucanr.edu/home-and-landscape/brown-rot-blossom-and-twig-blight/ and https://ipm.ucanr.edu/PMG/GARDEN/FRUIT/DISEASE/brownrot.html

[UCB] UC IPM Pest Management Guidelines: Peach - Brown rot blossom and twig blight (Monilinia biology; same pathogens as on cherry). https://ipm.ucanr.edu/agriculture/peach/brown-rot-blossom-and-twig-blight/

[UCR] UC IPM Pest Management Guidelines: Prune - Brown rot blossom and twig blight. https://ipm.ucanr.edu/agriculture/prune/brown-rot-blossom-and-twig-blight

[MB] Michigan State University IPM - Brown rot. https://www.canr.msu.edu/ipm/diseases/brown_rot

[NEB] New England Tree Fruit Network - Cherries: brown rot, bacterial canker and cherry leaf spot. https://netreefruit.org/stone-fruit/cherries/diseases/brown-rot and https://netreefruit.org/book/export/html/335

[MPR] MSU Extension - Pre-harvest disease control in cherry. https://www.canr.msu.edu/news/pre_harvest_disease_control_in_cherry

[CT] Connecticut Agricultural Experiment Station - Disease control for home cherry orchards. https://portal.ct.gov/CAES/Fact-Sheets/Plant-Pathology/Disease-Control-for-Home-Cherry-Orchards

[EPA] US EPA-accepted label, Captan 50-WP (1993), compatibility and plant-safety statements. https://www3.epa.gov/pesticides/chem_search/ppls/010182-00145-19930607.pdf

2. Bacterial Canker (Gummosis)

Cherry | Pseudomonas syringae (pv. syringae)

Crop	Cherry (Prunus avium)
Disease	Bacterial Canker (Gummosis)
Pathogen	Pseudomonas syringae (pv. syringae)
Type	Bacterial (plant-surface-, rain- and wound-borne)

At a glance. Bacterial disease of buds, blossoms, spurs and bark that girdles branches and can kill young trees. No highly effective bactericide exists; management rests on avoiding stress, wounds and wet canopies.

1. Identification

Symptoms

- Crop / stage: young and mature sweet cherry trees; buds, blossoms, spurs, shoots, trunk.
- Cankers on twigs at the base of flower and leaf buds, in pruning wounds or at the base of spurs, exuding amber-coloured gum (N1).
- Cankers spread upwards and form sunken areas in winter. Infected dormant buds may be killed or may open normally in spring and collapse in early summer; infected buds can also be symptomless (N1).
- Other symptoms: blossom blast, fruit spots, leaf spots and shot-holes, dead buds, and branch and trunk cankers, sometimes with gumming (MBC). Cankers can girdle and kill branches and sometimes entire trees (UCD).

Distinguishing features

- Blossom death and shoot dieback look like brown rot blight. Shoots killed by bacterial blast generally turn black, whereas brown rot shoots turn brown to tan (UCH). Brown rot can also be confused with bacterial canker (NEB).
- Silver leaf (Chondrostereum purpureum) also causes twig and branch die-back; confirm by cutting a branch at least 2.5 cm thick, wetting the cut surface and looking for a brown stain in the wood (N1).

2. Causes and progression

- Pathogen: Pseudomonas syringae (pv. syringae) (MBC). The Indian IPM package names it only as Pseudomonas sp. (N1).
- Survival and spread: the bacterium survives on plant surfaces and is spread by splashing rain (N1). Under suitable conditions it can colonise and move systemically within the tree (MBC).
- Who is affected: young trees, about 2 to 8 years old, are most affected; the disease rarely occurs in the first year after planting and is uncommon in nurseries; vigorous trees are less susceptible, and the disease is worse in low or sandy spots (N1). In California it is usually most severe in orchards 1 to 6 years old (UCD).

3. Favourable conditions

- Cold, wet conditions in stressed trees: low nutrient availability, nematode feeding, poor soil or mismanaged irrigation (UCD). The IPM package also states that high moisture and low spring temperatures favour the bacterium (N1); the same package elsewhere lists high humidity with warm temperature, so the two statements differ.
- Excess nitrogen, especially late in the season, promotes late growth that is prone to low-temperature injury in early winter, followed by bacterial infection (PUR).
- Most sweet cherry cultivars are susceptible or highly susceptible (MBC, PUR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use healthy planting material and discard nursery stock that shows any sign of bacterial canker (NEB, N1).
- Do not interplant new trees among old ones; old infected trees are a major source of inoculum (PUR).

- For the first 2 to 3 years keep irrigation water off the above-ground parts of the tree; consider withholding water in late summer so trees harden off (PUR).
- Avoid excess nitrogen, especially late in the season (PUR); the IPM package advises balanced nutrition with nitrogen in split doses (N1).
- Pruning time affects canker occurrence (MBC). Advice differs by region: some studies show less canker when pruning is delayed until spring (PUR), while a New England resource advises pruning immediately after harvest (NEB). A tested timing for Indian temperate orchards was not found; ask the local SAU / KVK.
- Choose a cherry variety and rootstock suited to the site so the tree is not stressed (N1); whitewash trunks with a glue-based wash to avoid sunburn and winter injury, and keep trees free of mechanical wounds, winter injury, crotch separation and cankers (N1).
- Collect and destroy diseased plant parts (N1).

Regional note (temperate India)

No state-specific bacterial canker recommendation (SKUAST-K, YSPUHF Nauni, ICAR-CITH) was found in the sources reviewed.

5. Remedies and treatment

- Highly effective management strategies are not available. The main tactics described by Michigan State University are copper to reduce P. syringae populations during bloom before frosts, and correctly timed pruning (MBC).

Chemical options with verification status

No product verified (tag --).

No product verified for cherry bacterial canker / gummosis. No cherry or bacterial-canker entry for any copper or antibiotic product was found in the CIB&RC fungicide compilation reviewed; the Indian IPM package gives no chemical for this disease (N1).

- Copper is described as a main tactic in Michigan (MBC), but extension advice for sweet cherry is not uniform: a New England guide advises against copper on sweet cherry (NEG). Because no Indian cherry registration was found, copper is not recommended here.
- Copper oxychloride and streptomycin-tetracycline entries in the CIB&RC list relate to other crops (for example citrus, chillies, apple fire blight), not cherry (CIB).

6. Sources and citations

[N1] AESA based IPM package for Cherry. Directorate of Plant Protection, Quarantine & Storage (DPPQS) and National Institute of Plant Health Management (NIPHM), Ministry of Agriculture & Farmers Welfare, Government of India. https://niphm.gov.in/IPMPackages/Cherry.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS - Major Uses of Pesticides: Fungicides (registered under the Insecticides Act, 1968), up to 30/09/2025. Copy reviewed: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[UCD] UC Davis Fruit & Nut Research and Information Center - Cherry diseases and disorders. https://fruitsandnuts.ucdavis.edu/node/5006

[UCH] UC IPM - Brown rot blossom and twig blight (home and landscape; stone fruit). https://ipm.ucanr.edu/home-and-landscape/brown-rot-blossom-and-twig-blight/ and https://ipm.ucanr.edu/PMG/GARDEN/FRUIT/DISEASE/brownrot.html

[NEB] New England Tree Fruit Network - Cherries: brown rot, bacterial canker and cherry leaf spot. https://netreefruit.org/stone-fruit/cherries/diseases/brown-rot and https://netreefruit.org/book/export/html/335

[NEG] 2011 New England Tree Fruit Management Guide, Ch. 13 - General pest management considerations: cherries (UMass Amherst). https://ag.umass.edu/sites/ag.umass.edu/files/pdf-doc-ppt/2011-13-cherry.pdf

[MBC] MSU Extension - Managing bacterial canker in sweet cherries: what are the options? https://www.canr.msu.edu/news/managing_bacterial_canker_in_sweet_cherries_what_are_the_options

[PUR] Purdue University, Facts for Fancy Fruit - Ten step program to manage bacterial canker of sweet cherry. https://fff.hort.purdue.edu/?p=2330

3. Cherry Leaf Spot (Shot Hole)

Cherry | Blumeriella jaapii in the international literature; Cercospora circumcissa / C. rubrotincta in the Indian IPM package

Crop	Cherry (Prunus avium)
Disease	Cherry Leaf Spot (Shot Hole)
Pathogen	Blumeriella jaapii in the international literature; Cercospora circumcissa / C. rubrotincta in the Indian IPM package
Type	Fungal (leaf-debris- and soil-borne)

At a glance. Leaf disease that causes spotting, shot-holing and early leaf drop, weakening trees over successive seasons. Destroying fallen leaves plus protective sprays is the standard approach.

1. Identification

Symptoms

- Crop / stage: cherry leaves from spring through the season; fruit, stems and petioles in severe cases.
- Brown spots on leaves that gradually expand with a reddish-brown margin, on both leaf surfaces; with many spots the leaves look burnt (N1).
- Small purple spots on the upper leaf surface; the centres of older spots can drop out to leave a shot-hole. Spots on sweet cherry are often larger and more circular than on sour cherry (CT).
- On the underside of the leaf, white spore masses can be seen under the lesions (MFS).
- Fruit, stems and petioles can be infected when the disease is severe (CT). Severe early leaf drop interferes with fruit ripening and colour, and trees with poor reserves are prone to winter injury and dieback (WVU); leaf spot also reduces flower-bud survival and fruit set (CT).

Distinguishing features

- Coryneum blight / shot hole (Stigmina carpophila) starts as small red spots on young leaves that enlarge, turn purple with a tan-white centre and drop out; fruit can show purple-red spots (N1).
- Bacterial canker can also cause leaf spots and shot-holes (MBC).
- Pathogen identity: the Indian IPM package names Cercospora species, whereas the international literature names Blumeriella jaapii (formerly Coccomyces hiemalis) for cherry leaf spot / shot hole (CT, N1). Confirm the pathogen locally before relying on pathogen-specific information.

2. Causes and progression

- Pathogen: see 'Distinguishing features' on the naming difference (N1, CT).
- Survival: in soil and plant debris (N1); the leaf spot fungus overwinters in fallen leaves infected the previous season (UAX).
- Progression: spores from overwintered leaves start infections on new foliage; spore discharge is highest across roughly 16-29 C (60-85 F) and lowest at about 5-8 C (41-46 F) (MFS; temperatures converted from degrees Fahrenheit).

3. Favourable conditions

- Lack of air movement, too-wet soil, too much shade or too much sun, and lack of nitrogen and potassium (N1).
- Wet seasons raise disease pressure and call for a more aggressive spray schedule on tart cherry (NEB); warming temperatures after shuck split favour spore discharge and infection (MFS).
- Sweet cherry is generally less susceptible than tart cherry, though still needs protection (NEB, WVU).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Collect, remove and burn or otherwise dispose of infected leaves (N1). Rake or chop fallen leaves (WVU); good sanitation greatly reduces incidence in small plantings (CT).

- Keep trees vigorous with balanced nutrition, including adequate nitrogen and potassium; avoid shade, water-logging and poor air movement (N1).
- Avoid overhead sprinklers or wetting the foliage (N1, in the shot hole entry).
- For resistant or tolerant varieties the IPM package refers growers to the nearest KVK / SAU / ICAR institute (N1).

Regional note (temperate India)

No state-specific leaf spot spray schedule recommendation (SKUAST-K, YSPUHF Nauni, ICAR-CITH) was found in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Lime sulphur 22% SC	Cherry - leaf spot	2-5 L formulation/ha (a.i. printed as 0.44-1.10 'per ha')	Not stated	V1	CIB&RC; 3 applications: after petal fall, 2 weeks later and after harvest. Cross-check (N1): 0.8-2.0 L per acre.
Zineb 75% WP	Cherries - leaf spot	1.5-2 kg formulation/ha (1.125-1.5 kg a.i./ha)	750-1000 L/ha	V1	CIB&RC; no waiting period stated. Confirm that the registration is still current.
Captan 50% WP	Leaf spot / shot hole	1 kg per acre	300-400 L per acre	V2	N1 only. CIB&RC lists captan on cherry for brown rot, not leaf spot.
Captan 75% WP	Leaf spot / shot hole	0.67 kg per acre	See note	V2	N1 only. The package prints the water volume as 6-8 L per acre, which looks like a misprint, so it is not reproduced; follow the product label.

V1 - Verified against CIB&RC. The product is listed for CHERRY and this disease in the CIB&RC 'Major Uses of Pesticides - Fungicides' compilation (up to 30/09/2025). The dose is reproduced as printed there.
V2 - Government / university recommendation only. Published in the Government of India NIPHM-DPPQS AESA IPM package for cherry (N1). The cherry label claim for this disease could NOT be matched in the CIB&RC compilation, so confirm the current CIB&RC registration and product label before use.

- Do not mix captan with lime sulphur; alkaline materials such as lime sulphur reduce captan's activity (EPA label statement, US).
- A leaf spot programme that works in one region depends on timing against local infection periods. None of the sources reviewed gives an Indian temperate-zone timing, so use the label and the local SAU / KVK schedule.

6. Sources and citations

[N1] AESA based IPM package for Cherry. Directorate of Plant Protection, Quarantine & Storage (DPPQS) and National Institute of Plant Health Management (NIPHM), Ministry of Agriculture & Farmers Welfare, Government of India. https://niphm.gov.in/IPMPackages/Cherry.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS - Major Uses of Pesticides: Fungicides (registered under the Insecticides Act, 1968), up to 30/09/2025. Copy reviewed: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[NEB] New England Tree Fruit Network - Cherries: brown rot, bacterial canker and cherry leaf spot. https://netreefruit.org/stone-fruit/cherries/diseases/brown-rot and https://netreefruit.org/book/export/html/335

[MBC] MSU Extension - Managing bacterial canker in sweet cherries: what are the options? https://www.canr.msu.edu/news/managing_bacterial_canker_in_sweet_cherries_what_are_the_options

[CT] Connecticut Agricultural Experiment Station - Disease control for home cherry orchards. https://portal.ct.gov/CAES/Fact-Sheets/Plant-Pathology/Disease-Control-for-Home-Cherry-Orchards

[MFS] MSU Extension - Fungicide considerations for cherry leaf spot control at first cover.
https://www.canr.msu.edu/news/fungicide_considerations_for_cherry_leaf_spot_control_at_first_cover

[UAX] University of Arkansas Cooperative Extension - Cherry leaf spot.
https://www.uaex.uada.edu/yard-garden/plant-health-clinic/disease-notes/posts/cherry-leaf-spot.aspx

[WVU] West Virginia University Extension - Cherry leaf spot.
https://extension.wvu.edu/lawn-gardening-pests/plant-disease/tree-fruit-disease/cherry-leaf-spot

[EPA] US EPA-accepted label, Captan 50-WP (1993), compatibility and plant-safety statements.
https://www3.epa.gov/pesticides/chem_search/ppls/010182-00145-19930607.pdf

4. Powdery Mildew

Cherry | Podosphaera clandestina

Crop	Cherry (Prunus avium)
Disease	Powdery Mildew
Pathogen	Podosphaera clandestina
Type	Fungal (obligate parasite; airborne conidia)

At a glance. White powdery growth on leaves, shoot tips and fruit. Young leaves and shoots are affected first, and fruit are most vulnerable up to pit hardening. Air circulation, sanitation and scouting are the main tools sourced here.

1. Identification

Symptoms

- Crop / stage: young leaves, shoot tips and fruit of cherry.
- Light-coloured circular lesions on leaves, often inside the canopy, that develop a powdery appearance and may coalesce to cover the leaf (N1). White powdery growth is typically most visible on the underside of leaves (WSU).
- Severely infected leaves blister or curl upward and may drop; infected shoots can be distorted and stunted (N1, MPM).
- Fruit: depressed, roughly circular lesions that may or may not show visible growth; fruit infected when young can be deformed (N1, MPM, BCT).
- Late in the season, small black fruiting bodies (cleistothecia / chasmothecia) appear in older colonies (MPM, BCT).

Distinguishing features

- Powdery mildew is first seen on the underside of leaves, so scouting under the leaf is needed for early detection (OMA).
- Spray residue can look like mildew but is denser and more visible (OMA).
- Damage is described as greater on sour cherry than on sweet cherry (MPM).

2. Causes and progression

- Pathogen: Podosphaera clandestina (N1, UCM).
- Survival: sources differ on where it overwinters. The Indian IPM package and UC IPM state it overwinters in buds on twigs and as chasmothecia on bark (N1, UCM); a British Columbia guide reports no evidence of survival as mycelium in dormant buds and places overwintering as cleistothecia in leaf litter and bark crevices (BCT).
- Spread: secondary spores produced in spring spread the fungus to new growth (N1); conidia are spread by wind (OMA).

3. Favourable conditions

- Low rainfall, high humidity and warm temperatures of about 21-27 C (70-80 F) (N1, UCM; converted from degrees Fahrenheit).
- Other guides give 15-25 C with humid conditions (BCT), or high humidity, warm days and cool nights (WSU).
- Fruit are most susceptible up to the pit-hardening stage (BCT).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Keep proper spacing and prune so the canopy has good air circulation and lower humidity (N1, WSU).

- Provide full sun, avoid heavy nitrogen that promotes succulent shoots, rake and remove fallen leaves, avoid overhead watering while fruit develop, and prune out and destroy severely infected shoots and infected suckers (WSU).
- Scout early: check leaves on suckers and spurs arising from the lower main limbs, from early bloom (USU).
- Susceptibility differs by cultivar; cultivars such as Black Tartarian, Bing, Rainier and Lapins are reported as highly susceptible in Utah (USU). For resistant or tolerant varieties in India the IPM package refers growers to the nearest KVK / SAU / ICAR institute (N1).

Regional note (temperate India)

No state-specific powdery mildew recommendation (SKUAST-K, YSPUHF Nauni, ICAR-CITH) was found in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

No product verified (tag --).

No product verified for cherry powdery mildew. The CIB&RC fungicide compilation reviewed lists powdery mildew products for apple, grape, mango and other crops, but no cherry powdery mildew entry; the Indian IPM package gives no chemical for it (N1, CIB).

- UC IPM advises protectant sprays soon after petal fall in California (UCM), but none of the products there could be matched to cherry powdery mildew in India, so none are reproduced. Fungicide registrations on other crops are not extended to cherry.
- For local options, check with the nearest KVK or the state agricultural / horticultural university.

6. Sources and citations

[N1] AESA based IPM package for Cherry. Directorate of Plant Protection, Quarantine & Storage (DPPQS) and National Institute of Plant Health Management (NIPHM), Ministry of Agriculture & Farmers Welfare, Government of India. https://niphm.gov.in/IPMPackages/Cherry.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS - Major Uses of Pesticides: Fungicides (registered under the Insecticides Act, 1968), up to 30/09/2025. Copy reviewed: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[UCM] UC IPM Pest Management Guidelines: Cherry - Powdery mildew (Podosphaera clandestina). https://ipm.ucanr.edu/PMG/r105100411.html

[MPM] Michigan State University IPM - Powdery mildew of cherry. https://www.canr.msu.edu/ipm/diseases/powdery_mildew_of_cherry

[WSU] Washington State University HortSense - Cherry: powdery mildew. https://hortsense.cahnrs.wsu.edu/fact-sheet/cherry-powdery-mildew/

[BCT] BC Tree Fruit Production Guide - Powdery mildew (sweet cherries). https://www.bctfpg.ca/pest_guide/info/101

[USU] Utah State University Extension - Powdery mildew of fruit. https://extension.usu.edu/pests/ipm/notes_ag/fruit-powdery-mildew

[OMA] OMAFRA (Ontario) - Powdery mildew of tart cherry. https://www.omafra.gov.on.ca/IPM/english/tender/diseases-and-disorders/powdery-tc.html

5. Crown Gall

Cherry | Agrobacterium tumefaciens (also named Rhizobium radiobacter)

Crop	Cherry (Prunus avium)
Disease	Crown Gall
Pathogen	Agrobacterium tumefaciens (also named Rhizobium radiobacter)
Type	Bacterial (soil- and nursery-stock-borne; wound entry)

At a glance. Soil-borne bacterial disease that causes galls on roots and the crown, stunting young trees. It enters through wounds and is often introduced on infected nursery stock. Prevention is the only reliable approach.

1. Identification

Symptoms

- Crop / stage: nursery trees and young orchards (also older trees).
- Rough, abnormal galls on roots or the trunk base; the Indian IPM package describes them as soft and spongy, with the centres of older galls decaying (N1). Young galls are light in colour and become dark and hard with age (PSU).
- Galls are often at or just below the soil surface, on the root crown (PNR); size ranges from under 1 cm to about 10 cm (MCG).
- Young trees become stunted; older trees often develop secondary wood rots (N1). Heavily galled trees may show reduced growth and nutritional deficiency symptoms because water and nutrient flow is disrupted (PSU).

Distinguishing features

- Any root or rootstock/scion-union disorder can produce similar above-ground symptoms; the presence of galls separates crown gall from them (MCG).
- A gall can be hard to tell from callus growth, especially at the graft or bud union (PNR).

2. Causes and progression

- Pathogen: Agrobacterium tumefaciens, now also called Rhizobium radiobacter (N1, PCG).
- Survival: the bacterium survives in soil for several years, often spreading from diseased nursery stock (PCG, N1). Its numbers do not fall quickly after infected trees are removed, so it can persist for years (MCG).
- Entry: through wounds, whether from planting, tillage equipment, pruning, grafting, chewing insects or nematodes (PSU, PNR).
- Progression: infection stimulates abnormal tissue growth; symptoms may not appear for over a year if infection occurs at low temperature (below about 15 C / 59 F) (PNR; converted from degrees Fahrenheit).

3. Favourable conditions

- Wounds at the crown and roots (PSU).
- Planting infected nursery stock, or planting in soil with a recent history of crown gall (PCG).
- The Indian IPM package lists high humidity and warm temperature as favouring the disease (N1).
- Rootstock susceptibility differs: Colt cherry rootstock is reportedly quite susceptible, while Gisela 5 and 6 may have useful resistance in contaminated soil (PCG; Oregon handbook, not an Indian recommendation).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use only pathogen-free nursery stock, inspect roots and crowns, and do not plant any tree with galls (PCG, MCG). The IPM package also says to avoid planting saplings infected with disease (N1).
- Do not plant in soil with a recent history of crown gall, and avoid replanting a susceptible host on the same site after removing infected trees (PCG, MCG).

- Avoid injury to bark and roots at planting and cultivation; avoid sites heavily infested with root-attacking insects and nematodes; disinfect pruning tools; do not plant deeper than the trees grew in the nursery (PCG, PNN).
- Pre-plant soil solarization has been effective for nursery stock in western Oregon (PCG); the Indian package describes covering beds with 45-gauge (0.45 mm) polythene for three weeks before sowing to reduce soil-borne problems (N1).
- Remove and destroy seriously declining trees with large galls around the crown (PCG).

Regional note (temperate India)

No state-specific crown gall recommendation (SKUAST-K, YSPUHF Nauni, ICAR-CITH) was found in the sources reviewed.

5. Remedies and treatment

- Once a tree is infected, there is no curative treatment in the sources reviewed. Management is preventive.

Chemical options with verification status

No product verified (tag --).

No product verified for cherry crown gall. No cherry crown gall entry in the CIB&RC fungicide compilation reviewed, and the Indian IPM package gives no chemical (N1, CIB). In the US, no registered chemicals are reported to control crown gall effectively (PNN).

- A biological preventive (Agrobacterium radiobacter strain 84) is used abroad on nursery stock; it is preventive only and must be applied a few hours after wounding (PCG). Its registration status in India was not verified, so it is not recommended here.

6. Sources and citations

[N1] AESA based IPM package for Cherry. Directorate of Plant Protection, Quarantine & Storage (DPPQS) and National Institute of Plant Health Management (NIPHM), Ministry of Agriculture & Farmers Welfare, Government of India. https://niphm.gov.in/IPMPackages/Cherry.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS - Major Uses of Pesticides: Fungicides (registered under the Insecticides Act, 1968), up to 30/09/2025. Copy reviewed: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[PCG] Pacific Northwest Plant Disease Management Handbook - Cherry (Prunus spp.): crown gall. https://pnwhandbooks.org/plantdisease/host-disease/cherry-prunus-spp-crown-gall

[PNN] Pacific Northwest Plant Disease Management Handbook - Crown gall in nursery stock. https://pnwhandbooks.org/node/409

[PNR] Pacific Northwest Plant Disease Management Handbook - Crown gall (general biology, rose entry). https://pnwhandbooks.org/plantdisease/host-disease/rose-rosa-spp-hybrids-crown-gall

[MCG] Michigan State University IPM - Crown gall. https://www.canr.msu.edu/ipm/diseases/crown_gall

[PSU] Penn State Extension - Crown gall on stone fruit in home plantings. https://extension.psu.edu/crown-gall-on-stone-fruit-in-home-plantings

6. Collar Rot / Root and Crown Rot (Phytophthora)

Cherry | Phytophthora spp.

Crop	Cherry (Prunus avium)
Disease	Collar Rot / Root and Crown Rot (Phytophthora)
Pathogen	Phytophthora spp.
Type	Oomycete (fungus-like), soil- and water-borne

At a glance. Soil-borne rot of the root crown and roots that kills young trees quickly after the first warm spring weather. It is driven by waterlogged soil, so site choice, drainage and irrigation management are the main controls.

1. Identification

Symptoms

- Crop / stage: young cherry trees especially; roots, crown and lower trunk.
- Symptoms depend on how much of the root or crown tissue is affected and how fast it is destroyed. Crown rots advance rapidly, and trees collapse and die soon after the first warm weather of spring; leaves wilt, dry and stay attached (N1, UCP).
- Chronic root infections cause reduced growth, early senescence and leaf fall; such trees may be unthrifty for several years before dying (N1).
- Darkened areas in the bark around the crown and upper roots (UCQ); beads of sap on the trunk and cracking can occur (UCG).
- Phytophthora typically kills young trees because their root systems and crowns are small (N1, UCP).

Distinguishing features

- Affected plants wilt even when well watered; unlike plants under water stress, they do not recover when irrigated (UCQ).
- White root rot (Dematophora necatrix) shows yellow foliage, shrivelled fruit and cottony white mycelium on small feeder roots, with older mycelium turning grey or black (N1).
- Verify that Phytophthora is the cause before treating, because a number of factors or pest problems can cause poor growth and tree death (UCP, N1).

2. Causes and progression

- Pathogen: Phytophthora sp. (N1); soil-borne (UCP).
- Survival: the pathogen overwinters in infected trees or in colonised fallen plant debris (N1).
- Progression: infection of roots and crown in wet soil leads to rot of crown and root tissues, then wilting and collapse; symptoms are most often seen in spring and early summer when water demand is high (UCG, UCQ).

3. Favourable conditions

- Cool (10-16 C), wet soil (N1).
- Periods of 24 hours or more of saturated soil (UCP); over-irrigated soils (UCD).
- Rootstocks differ in susceptibility to different Phytophthora species; none is resistant to all (UCP, N1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant in well-drained soil to reduce the frequency and duration of water-saturated soil (N1).
- Avoid sites with a history of Phytophthora root and crown rot, especially with susceptible rootstocks (UCP, N1).
- Plant new orchards on berms or mounds to improve drainage at the crown, and design irrigation so that the trunk and crown are not wetted by sprinklers (UCP, N1). More frequent but shorter irrigations reduce the risk (UCP).

- Do not cover the graft union with soil and do not water the crown directly (UCG).
- If crown rot is suspected, carefully cut away affected bark at the soil line; trees can sometimes be saved by removing soil from the base down to the tops of the main roots and letting the crown tissue dry out (UCG).
- Indian IPM package: pits of 1 m x 1 m x 1 m filled with topsoil, 35-40 kg farmyard manure and 0.5 kg superphosphate one month before planting; neem cake at 100 kg per acre at transplanting to reduce soil-borne diseases in nurseries (N1).

Regional note (temperate India)

No state-specific collar rot recommendation (SKUAST-K, YSPUHF Nauni, ICAR-CITH) was found in the sources reviewed.

5. Remedies and treatment

- The Indian IPM package lists only cultural and mechanical measures for collar / root rot and no chemical (N1). Uproot and dispose of badly infected trees (N1, white root rot entry).

Chemical options with verification status

No product verified (tag --).

No product verified for cherry collar rot / root rot. No cherry Phytophthora entry in the CIB&RC fungicide compilation reviewed (metalaxyl-containing and other Phytophthora entries there are for crops such as citrus, black pepper, potato and ginger, not cherry) (CIB); the IPM package gives no chemical (N1).

- UC IPM guidance lists fungicides for cherry Phytophthora in California (UCP). None could be matched to a cherry registration in India, so none are reproduced.

6. Sources and citations

[N1] AESA based IPM package for Cherry. Directorate of Plant Protection, Quarantine & Storage (DPPQS) and National Institute of Plant Health Management (NIPHM), Ministry of Agriculture & Farmers Welfare, Government of India. https://niphm.gov.in/IPMPackages/Cherry.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS - Major Uses of Pesticides: Fungicides (registered under the Insecticides Act, 1968), up to 30/09/2025. Copy reviewed: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[UCD] UC Davis Fruit & Nut Research and Information Center - Cherry diseases and disorders. https://fruitsandnuts.ucdavis.edu/node/5006

[UCP] UC IPM Pest Management Guidelines: Cherry - Phytophthora root and crown rot. https://ipm.ucanr.edu/agriculture/cherry/phytophthora-root-and-crown-rot/

[UCQ] UC IPM Quick Tips - Phytophthora crown and root rot. https://ipm.ucanr.edu/legacy_assets/PDF/QT/qtphytophthora.pdf

[UCG] UC IPM (garden and landscape) - Phytophthora root and crown rot of fruit trees. https://ipm.ucanr.edu/PMG/GARDEN/FRUIT/DISEASE/phytortcrrot.html

Consolidated source list

[N1] AESA based IPM package for Cherry. Directorate of Plant Protection, Quarantine & Storage (DPPQS) and National Institute of Plant Health Management (NIPHM), Ministry of Agriculture & Farmers Welfare, Government of India. https://niphm.gov.in/IPMPackages/Cherry.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS - Major Uses of Pesticides: Fungicides (registered under the Insecticides Act, 1968), up to 30/09/2025. Copy reviewed: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf

[UCD] UC Davis Fruit & Nut Research and Information Center - Cherry diseases and disorders. https://fruitsandnuts.ucdavis.edu/node/5006

[UCH] UC IPM - Brown rot blossom and twig blight (home and landscape; stone fruit). https://ipm.ucanr.edu/home-and-landscape/brown-rot-blossom-and-twig-blight/ and https://ipm.ucanr.edu/PMG/GARDEN/FRUIT/DISEASE/brownrot.html

[UCB] UC IPM Pest Management Guidelines: Peach - Brown rot blossom and twig blight (Monilinia biology; same pathogens as on cherry). https://ipm.ucanr.edu/agriculture/peach/brown-rot-blossom-and-twig-blight/

[UCR] UC IPM Pest Management Guidelines: Prune - Brown rot blossom and twig blight. https://ipm.ucanr.edu/agriculture/prune/brown-rot-blossom-and-twig-blight

[MB] Michigan State University IPM - Brown rot. https://www.canr.msu.edu/ipm/diseases/brown_rot

[NEB] New England Tree Fruit Network - Cherries: brown rot, bacterial canker and cherry leaf spot. https://netreefruit.org/stone-fruit/cherries/diseases/brown-rot and https://netreefruit.org/book/export/html/335

[MPR] MSU Extension - Pre-harvest disease control in cherry. https://www.canr.msu.edu/news/pre_harvest_disease_control_in_cherry

[CT] Connecticut Agricultural Experiment Station - Disease control for home cherry orchards. https://portal.ct.gov/CAES/Fact-Sheets/Plant-Pathology/Disease-Control-for-Home-Cherry-Orchards

[EPA] US EPA-accepted label, Captan 50-WP (1993), compatibility and plant-safety statements. https://www3.epa.gov/pesticides/chem_search/ppls/010182-00145-19930607.pdf

[NEG] 2011 New England Tree Fruit Management Guide, Ch. 13 - General pest management considerations: cherries (UMass Amherst). https://ag.umass.edu/sites/ag.umass.edu/files/pdf-doc-ppt/2011-13-cherry.pdf

[MBC] MSU Extension - Managing bacterial canker in sweet cherries: what are the options? https://www.canr.msu.edu/news/managing_bacterial_canker_in_sweet_cherries_what_are_the_options

[PUR] Purdue University, Facts for Fancy Fruit - Ten step program to manage bacterial canker of sweet cherry. https://fff.hort.purdue.edu/?p=2330

[MFS] MSU Extension - Fungicide considerations for cherry leaf spot control at first cover. https://www.canr.msu.edu/news/fungicide_considerations_for_cherry_leaf_spot_control_at_first_cover

[UAX] University of Arkansas Cooperative Extension - Cherry leaf spot. https://www.uaex.uada.edu/yard-garden/plant-health-clinic/disease-notes/posts/cherry-leaf-spot.aspx

[WVU] West Virginia University Extension - Cherry leaf spot. https://extension.wvu.edu/lawn-gardening-pests/plant-disease/tree-fruit-disease/cherry-leaf-spot

[UCM] UC IPM Pest Management Guidelines: Cherry - Powdery mildew (Podosphaera clandestina). https://ipm.ucanr.edu/PMG/r105100411.html

[MPM] Michigan State University IPM - Powdery mildew of cherry. https://www.canr.msu.edu/ipm/diseases/powdery_mildew_of_cherry

[WSU] Washington State University HortSense - Cherry: powdery mildew. https://hortsense.cahnrs.wsu.edu/fact-sheet/cherry-powdery-mildew/

[BCT] BC Tree Fruit Production Guide - Powdery mildew (sweet cherries). https://www.bctfpg.ca/pest_guide/info/101

[USU] Utah State University Extension - Powdery mildew of fruit. https://extension.usu.edu/pests/ipm/notes_ag/fruit-powdery-mildew

[OMA] OMAFRA (Ontario) - Powdery mildew of tart cherry. https://www.omafra.gov.on.ca/IPM/english/tender/diseases-and-disorders/powdery-tc.html

[PCG] Pacific Northwest Plant Disease Management Handbook - Cherry (Prunus spp.): crown gall. https://pnwhandbooks.org/plantdisease/host-disease/cherry-prunus-spp-crown-gall

[PNN] Pacific Northwest Plant Disease Management Handbook - Crown gall in nursery stock. https://pnwhandbooks.org/node/409

[PNR] Pacific Northwest Plant Disease Management Handbook - Crown gall (general biology, rose entry). https://pnwhandbooks.org/plantdisease/host-disease/rose-rosa-spp-hybrids-crown-gall

[MCG] Michigan State University IPM - Crown gall. https://www.canr.msu.edu/ipm/diseases/crown_gall

[PSU] Penn State Extension - Crown gall on stone fruit in home plantings. https://extension.psu.edu/crown-gall-on-stone-fruit-in-home-plantings

[UCP] UC IPM Pest Management Guidelines: Cherry - Phytophthora root and crown rot. https://ipm.ucanr.edu/agriculture/cherry/phytophthora-root-and-crown-rot/

[UCQ] UC IPM Quick Tips - Phytophthora crown and root rot. https://ipm.ucanr.edu/legacy_assets/PDF/QT/qtphytophthora.pdf

[UCG] UC IPM (garden and landscape) - Phytophthora root and crown rot of fruit trees. https://ipm.ucanr.edu/PMG/GARDEN/FRUIT/DISEASE/phytortcrrot.html

Items checked but not included

- Coryneum blight / shot hole (Stigmina carpophila), silver leaf canker, white root rot (Dematophora necatrix), whisker rot (Rhizopus stolonifer) and Prunus necrotic ringspot virus: listed in the Indian IPM package (N1) but outside this six-disease set; they can be added as further chapters.
- X-disease (phytoplasma): described for California (UCD) and not found in the Indian sources reviewed.
- State university or KVK cherry disease spray schedules (SKUAST-K, YSPUHF Nauni, ICAR-CITH): none retrieved.
- A news report of an unidentified drying disease of cherry trees in the Baghi belt of upper Shimla: no cause was identified in the report, so it is not treated as a disease chapter.
- Fungicides used on cherry abroad (sterol-inhibitor, strobilurin, dicarboximide and similar groups) and bactericidal copper programmes: not reproduced, because no cherry entry for them was found in the CIB&RC compilation reviewed.
- CIB&RC fungicide combination products (compilation pp. 42-79): scanned, with no cherry entry found. The CIB&RC bio-fungicide compilation was not checked.
- Registration changes after 30/09/2025: not checked.
- Insect pests of cherry (stem borer, shot hole borer, aphids and others, listed in N1): outside the scope of a disease document.